

2^{31} to $2^{31} - 1 \rightarrow \boxed{C}$

for i in range(0, n): start (inclusive)

for i in range(0, n, 2): step

for i in range(n): end (exclusive)

n(4)

```

- - - *
- - * * *
- * * * *
* * * * *
- * * * *
- - * * *
- - - *
  
```

① - Non row

② - Non who in 1st row

What needs to be printed?

n(3)

```

- - *
- * *
* * * *
- * * *
- - *
  
```

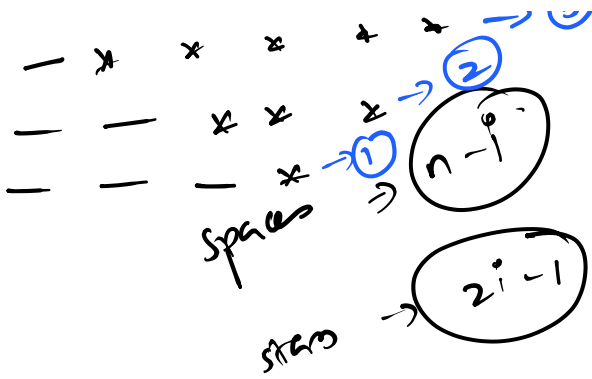
n(4)

```

- - - * → ①
- - * * * → ②
- * * * * → ③
* * * * * → ④
  
```

②

i	spaces	stars
1	3	1
2	2	3



3	1	5
4	0	7

$n=4$

0 $\rightarrow$ 1 3 5 7

1 $\rightarrow$ 3 5 7 1

2 $\rightarrow$ 5 7 1 3

3 $\rightarrow$ 7 1 3 5

$n=5$

1 3 5 7 9

3 5 7 9 1

5 7 9 1 3

7 9 1 3 5

9 1 3 5 7

0 $\rightarrow$ no. row $\rightarrow n$

1 $\rightarrow$ no. col $\rightarrow n$

$\downarrow$

What needs to be printed?

end $\rightarrow 2n-1$

start $\rightarrow 2i+1$

```
main.c
1 // Online C compiler to run C program online
2 #include <stdio.h>
3
4 int main() {
5     int arr[4];
6     printf("arr : %p\n", arr);
7     printf("0th : %p\n", &arr[0]);
8     printf("1st : %p\n", &arr[1]);
9     printf("2nd : %p\n", &arr[2]);
10    return 0;
11 }
```

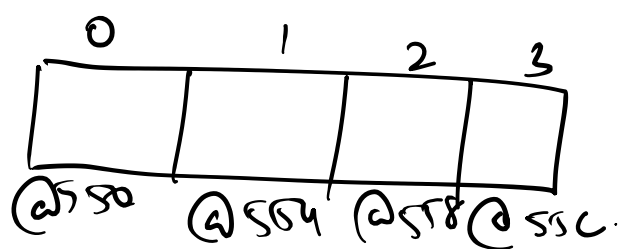
Output

```
arr : 0x7ffd59b6c550
0th : 0x7ffd59b6c550
1st : 0x7ffd59b6c554
2nd : 0x7ffd59b6c558
```

=== Code Execution Successful ===

0 550

arr



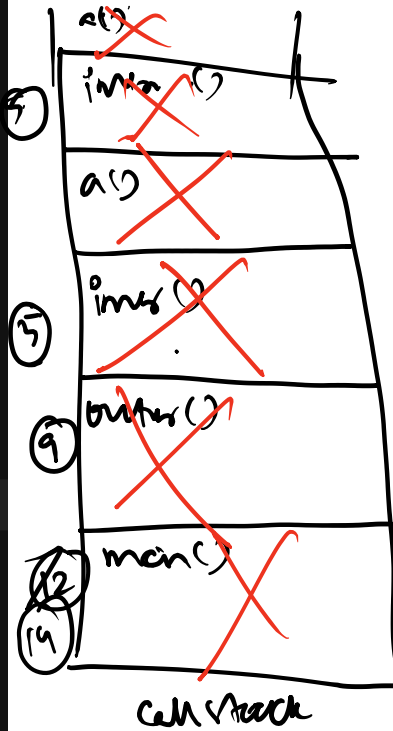
$arr[2] = \text{BAH}$

(2x4)

```

1  def a() :
2      print("Inside a ")
3
4  def inner() :
5      a()
6      print("Inside inner")
7
8  def outer() :
9      inner()
10     print("Inside outer")
11
12 outer()
13 print("Inside main")
14 inner()

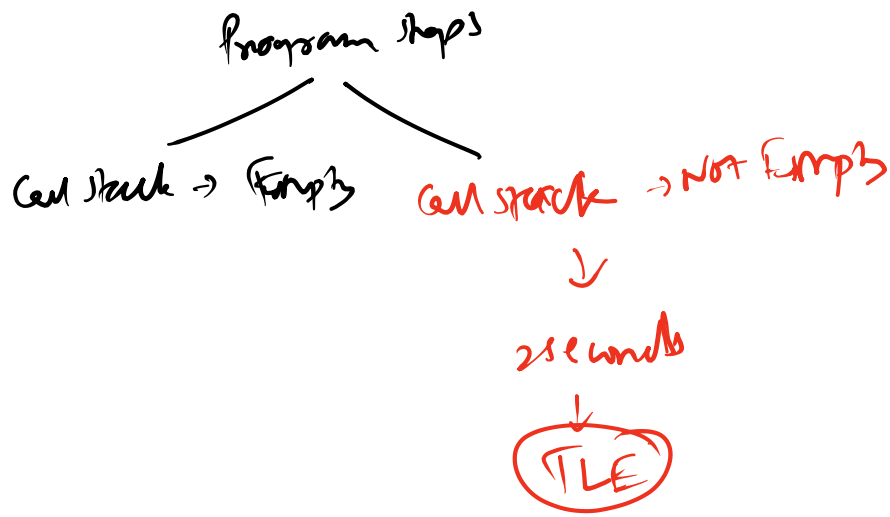
```



Console.

Inside a
Inside inner
Inside outer
Inside main
Inside a
Inside inner

→ Program stop
↓
Call stack → Empty



217. Contains Duplicate

Solved ✓

Easy

Topics

Companies

Given an integer array `nums`, return `true` if any value appears at least twice in the array, and return `false` if



appears at least twice in the array, and return false if every element is distinct.

Example 1:

Input: nums = [1, 2, 3, 1]

Output: true

2sec $\rightarrow 10^8$ 1sec $\rightarrow 10^7$ operations

Contains Duplicate	Two Sum
$T.C \rightarrow O(n^2)$ $(10^5)^2 \rightarrow 10^{10} \rightarrow 2.5 \text{ sec}$ <u>$2.5 > 2 \rightarrow \text{TLF}$</u>	$O(n^2)$ $(10^4)^2 \rightarrow 10^8 \rightarrow 2 \text{ sec}$

Set $\rightarrow$ stores elements (unique) $\Rightarrow$ no duplicate.



```
class Solution:
```

```
    def containsDuplicate(self, nums: List[int]) -> bool:
```

```
        seen = set()
```

```
        for i in range(len(nums)) :
```

```
            if nums[i] in seen :
```



```

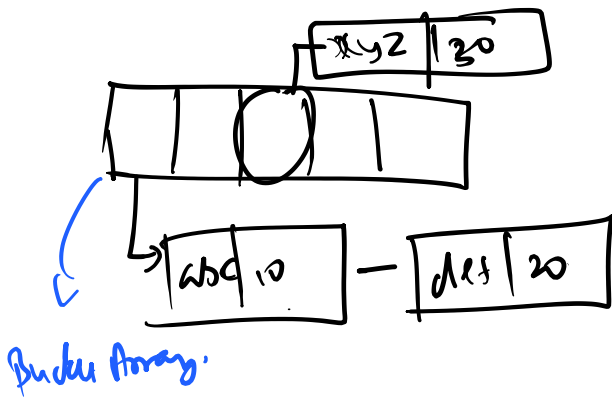
return True
seen.add(nums[i])
return False

```

↓
p.No: 217

↓
T.C $\rightarrow O(n)$

↓
S.C $\rightarrow O(n)$



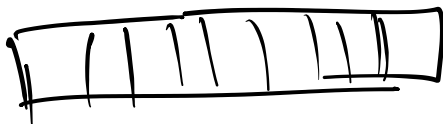
abc : 10 $\rightarrow$ Hashing $\rightarrow 0$
 def : 20 $\rightarrow 1$
 xyz : 30 $\rightarrow 2$

load factor $\rightarrow$

$$\gamma_b \geq 0.75$$

Enlarge the size $\leftarrow$ Bucket Array
 REHASH

$1/5 \rightarrow 0.2 < 0.75 \checkmark$
 $2/5 \rightarrow 0.4 < 0.75 \checkmark$
 $3/5 \rightarrow 0.6 < 0.75 \checkmark$
 $4/5 \rightarrow 0.8 > 0.75$



$4/10 \rightarrow 0.4 < 0.75$
 $5/10 \rightarrow 0.5 < 0.75$
 $6/10 \rightarrow 0.6 < 0.75$

↑ BASE

← REHASH

$7/10 \rightarrow 0.7 < 0.75$

$8/10 \rightarrow 0.8 > 0.75$

(KLE) - (8/10 → ...)

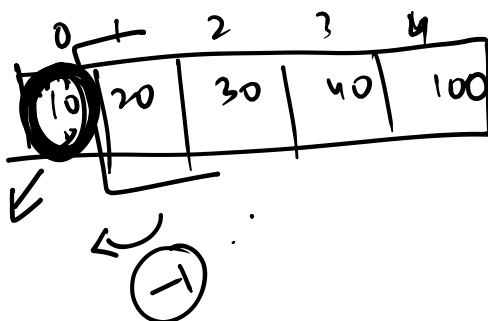
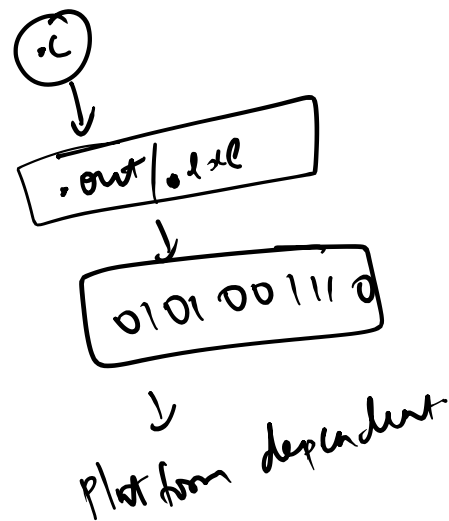
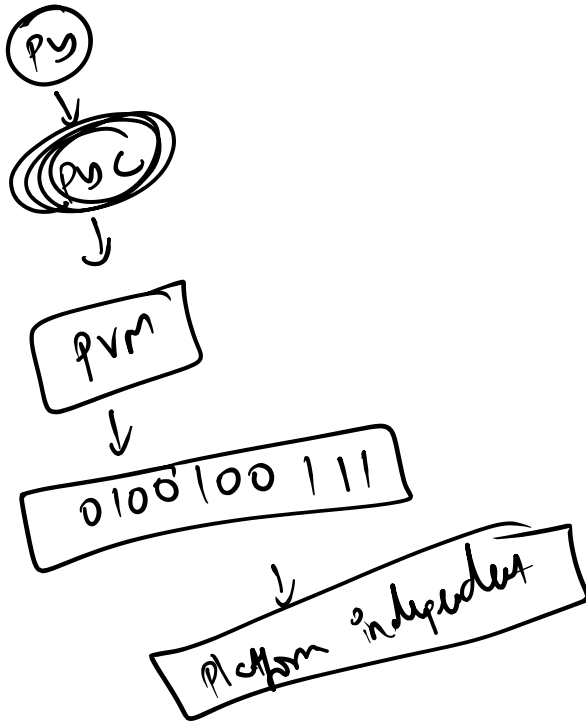
class Solution:

```
def twoSum(self, nums: List[int], target:
int) -> List[int]:
    visited = {}
    for i in range(len(nums)) :
        elem = target - nums[i]
        if elem in visited :
            j = visited[elem]
            return [j, i]
        visited[nums[i]] = i
```

→ p.No. ①

→ T.C → $O(n)$

→ S.C → $O(n)$



elem [10]

[0]